

StreetRace Manager Integration

Testing Report

Repository: <https://github.com/Athmeeya2006/dass-assignment-2>

1. Introduction

This report documents the integration testing of StreetRace Manager, a command-line system for underground street-race operations. The objective was to verify correct interaction and data flow across all required modules and the additional modules implemented for the assignment.

2. Modules Implemented

Required modules: Registration, Crew Management, Inventory, Race Management, Results, and Mission Planning.

Additional modules: Garage and Leaderboard.

3. Call Graph

Manual submission item: the assignment requires a hand-drawn call graph image. The repository already contains supporting material in `integration/diagrams/integration_call_graph.mmd` and `integration/diagrams/call_graph_description.md`. The final hand-drawn image should be placed in `integration/diagrams/` and attached to the submission package.

4. Integration Test Execution

Verified result: `python3 -m pytest integration/tests -q` produced **154 passed**.

The integration suite validates both module-to-module interaction and the underlying module-specific business rules that support those interactions.

5. Core Integration Scenarios

Scenario	Modules Involved	What Was Tested	Expected Result	Actual	Error
----------	------------------	-----------------	-----------------	--------	-------

Full race pipeline			Non-driver race		
ed	Registration, race management, results, inventory	Register a driver, add a car, create a race, assign both, start it, record the result, and verify that prize money is added to inventory cash.	Race completes and prize money is added correctly.	Pass	None
ed	Registration, race management, results, inventory	Attempt to	Operation is rejected with a	Pass	None

entry	race_	assign a mechanic as a race driver.	driver-related ValueError.	s	e
	mana				
	game				
Unregistered member race entry	nt	Attempt to assign a non-existent member to a race.			
	nt,		Lookup fails with KeyError.	Pas	Not on e
	regist			s	
Damaged car assignment	ration				
		Attempt to assign a damaged car to a race.	Operation is rejected with a damaged-car error.	Pas	Not on e
	ory			s	
Mission missing role					
	missi	Create a mission requiring a driver when	Mission assignment is rejected.	Pas	Not on e
	on_pl			a	
	annin			s	
	g,			s	
	crew_				
	mana				
	game				
	nt				

no
drive
r is
avail
able.

Assign a mission involving a damaged car without an available mechanic.

Damaged car mission needs mechanic

missi
on_pl
annin
g,
crew_
mana
geme
nt

Mission assignment is rejected with mechanic-related validation.

P
a
s
s

None

Assi
gn
and
com
plete
a
missi
on
with
all
requi
red
roles
prese
nt.

Mission becomes active, assigned members become unavailable, and all become available again after completion.

Missi
on
succes
s flow

regist
ration
,
crew_
mana
geme
nt,
missi
on_pl
annin
g

P
a
s
s

None

Driver
releas
ed

result
s,
crew_

Verif
y
drive

Driver is
unavailab
le during

P
a
s

N
o
n

		r			
		avail			
		abilit			
		y	race and		
after	mana	befor	available		
race	geme	e and	after	s	e
	nt	after	record_re		
		race	sult()).		
		com			
		pleti			
		on.			
		Run			
		two			
		races			
	regist	with			
	ration	diffe			
Leade	,	rent	Leaderbo		
rboard	race_	prize	ard sorts	P	N
reflect	mana	value	correctly	a	o
s	geme	s and	by wins	s	n
outco	nt,	verif	and	s	e
mes	result	y	earnings.		
	s,	order			
	leader	ing			
	board	on			
		leade			
		rboar			
		d.			
		Repa			
		ir a			
	invent	dama	Pre-repair		
Repai	ory,	ged	assignme		
r	garag	car	nt fails;	P	N
enable	e,	and	post-	a	o
s race	race_	then	repair	s	n
entry	mana	assig	assignme	s	e
	geme	n it	nt		
	nt	to a	succeeds.		
		race.			
Role	regist	Chan	Race	P	N

		ge a			
	ration	strate			
	,	gist	entry fails		
chang	crew_	role	before		
e	mana	to	role	a	o
enable	game	drive	change	s	n
s race	nt,	r and	and	s	e
entry	race_	retry	succeeds		
	mana	race	after it.		
	game	entry			
	nt	.			
		Ensu			
		re all			
		assig			
	missi	ned			
Missi	on_pl	crew	Assigned		
on	annin	are	members	P	N
compl	g,	relea	return to	a	o
etion	crew_	sed	available	s	n
frees	mana	after	state.	s	e
crew	game	missi			
	nt	on			
		com			
		pleti			
		on.			

6. Business Rules Verified

- A crew member must be registered before a role can be assigned or looked up.
- Only crew members with the driver role may be assigned to a race.
- Unavailable members cannot be reused incorrectly during race or mission flows.
- Damaged cars cannot be assigned to races until repaired.
- Race completion updates rankings, frees the driver, and updates inventory cash balance.
- Mission assignment checks all required roles before activation.
- Mission completion and failure correctly restore crew availability.
- Leaderboard output reflects combined results from race and mission systems.

7. Coverage Notes

In addition to the twelve high-value end-to-end scenarios, the repository contains dedicated test files for the registration, crew management, inventory, race management, results, mission planning,

leaderboard, and garage modules. Together, these contribute to the total of 154 passing integration-side tests.

8. Conclusion

The StreetRace Manager system integrates correctly across all required modules and both additional modules. The verified test results show that race workflows, mission workflows, repair flows, ranking updates, and role-based validation rules work together consistently.